Paperless-ngx on UGREEN NAS (Docker)

Guide (EN) – Version v1.0 – As of 27.12.2025

Community solution, not an official UGREEN product. Use at your own risk.

1. Introduction

Paperless-ngx is an open-source document management system (scans, OCR, tags, full-text search). This community package provides a ready-to-run Docker setup for UGOS (Paperless-ngx + PostgreSQL 18 + Redis 8) and adds an export script that creates regular ZIP backups and can optionally send email notifications.

2. Features

- Paperless-ngx container (latest) with persistent storage for data/media/consume/export.
- PostgreSQL 18 as database (named volume – avoids chmod/chown issues on first start).
- Redis 8 as broker/cache (named volume).
- Central configuration via one file: paperlessngx.env (Docker + export script).
- Export script: document_exporter -> versioned ZIP exports with retention.
- Optional SMTP email on failure and/or success (UTF-8 headers; optional ASCII icons).

3. Package contents and folder structure

Recommended location is a folder on a data volume, e.g. under /volume2/docker/. The directory structure in the package is already prepared.

Path	Description
/volume2/docker/paperless-ngx/docker-compose.yaml	Docker Compose for Paperless-ngx, PostgreSQL 18 and Redis 8
/volume2/docker/paperless-ngx/paperlessngx.env	Central configuration (passwords, DB, SMTP, export options)
/volume2/docker/paperless-ngx/scripts/paperless-export_with_mail.sh	Export script (ZIP exports + optional email)
/volume2/docker/paperless-ngx/data	Paperless data (configuration, index, etc.)
/volume2/docker/paperless-ngx/media	Documents/attachments
/volume2/docker/paperless-ngx/consume	Inbox folder (imports)
/volume2/docker/paperless-ngx/export	Export destination folder (ZIP backups)

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4. Requirements

- UGREEN NAS with UGOS and the Docker app installed (“Docker” / “Container”).
- Sufficient storage space on a data volume (recommended: /volume2).
- SMTP credentials for email notifications (server, port, user, password, sender, recipient).
- Optional: SSH access (for cron jobs / troubleshooting).

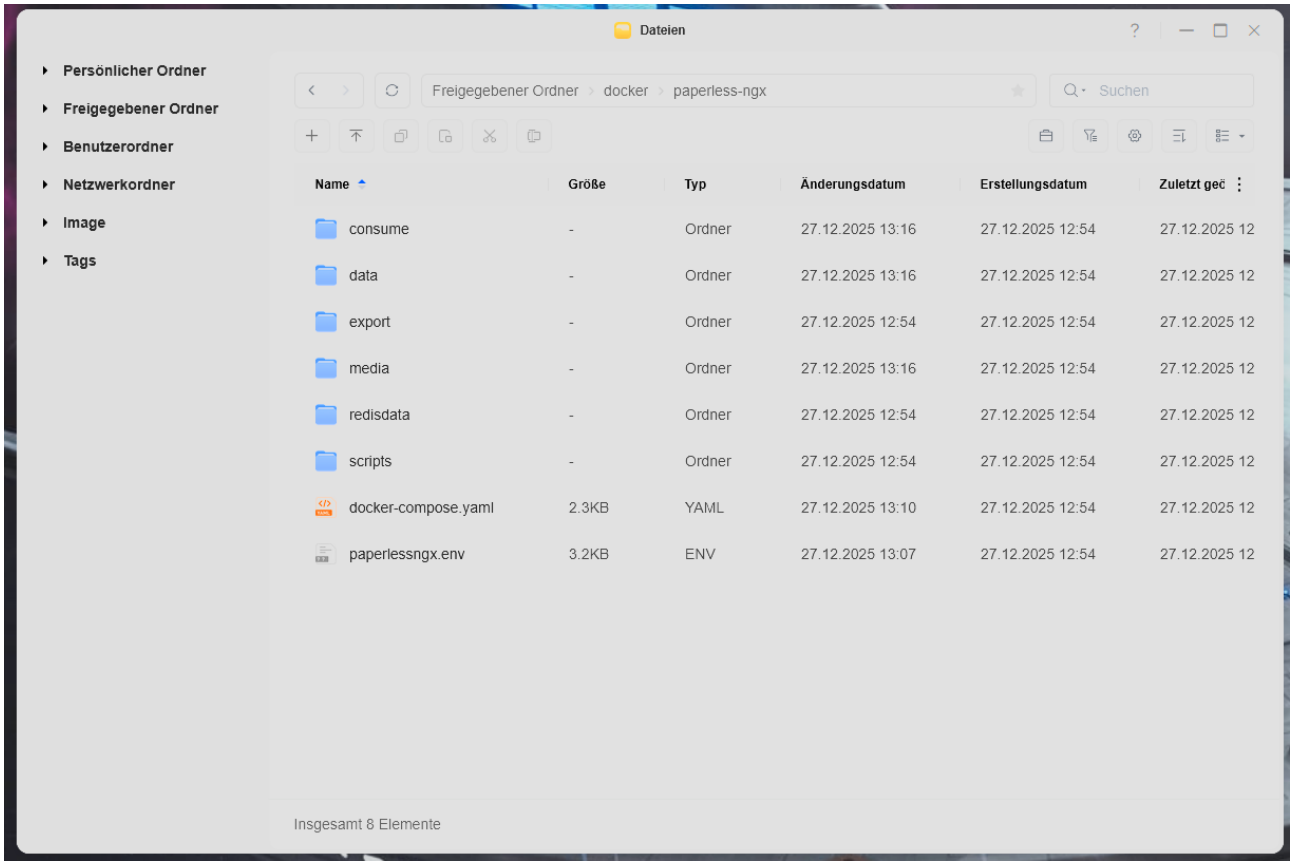
5. Installation (Docker Projects in UGOS)

5.1 Create folders and copy files

Example path:

```
mkdir -p /volume2/docker/paperless-ngx
cd /volume2/docker/paperless-ngx
```

Then copy the package files (docker-compose.yaml, paperlessngx.env and the scripts/ folder) into this directory. You can use UGOS “Files”, SMB or SCP. SSH note: if you log in via SSH and are not root, run `sudo -i` first before running commands.



5.2 Adjust paperlessngx.env

Open paperlessngx.env and adjust at least the following values:

- PAPERLESS_ADMIN_USER and PAPERLESS_ADMIN_PASSWORD (created only on the very first start).
- PAPERLESS_SECRET_KEY (set once and do NOT change later).
- POSTGRES_PASSWORD and PAPERLESS_DBPASS (must be identical).
- SMTP_* and MAIL_* (if email notifications are desired).

Tip: Generate a secret key:

```
openssl rand -base64 48
```

https://www.datenschutz.org/passwort-generator/or use https://www.datenschutz.org/passwort-generator/

Tip: Protect the ENV file (contains passwords):

```
chmod 600 /volume2/docker/paperless-ngx/paperlessngx.env
```

W:\paperless-ngx\paperlessngx.env - Notepad++

Datei Bearbeiten Suchen Ansicht Codierung Sprache Einstellungen Werkzeuge Makros Ausführen Plugins Fenster ?

paperlessngx.env

```
# paperlessngx.env  (KEY=VALUE)
# ============================================================
# Community-Paket: Paperless-ngx + Export-Skript (UGREEN NAS)
# Autor: Roman Glos (Community)
#
# ✅ Diese Datei wird von docker-compose UND den Skripten genutzt.
#    (Du kannst sie später optional aufteilen in:
#     - paperlessngx.env (Docker)
#     - scripts.env (Skripte/SMTP) )
#
# WICHTIG:
# - Werte mit Sonderzeichen am besten in Anführungszeichen setzen:  PASS="abc!?_$"
# - Datei schützen: chmod 600 paperlessngx.env
# ============================================================

# -------------------------
# UID / GID (deine Werte)
# -------------------------
USERMAP_UID=1009
USERMAP_GID=10

# -------------------------
# Paperless Admin User
# (wird beim ersten Start angelegt; danach kannst du es auskommentieren)
# -------------------------
PAPERLESS_ADMIN_USER=[redacted]
PAPERLESS_ADMIN_PASSWORD=[redacted]

# -------------------------
# Paperless Secret Key (einmalig setzen, dann NIE wieder ändern!)
# Tipp: openssl rand -base64 48
# -------------------------
PAPERLESS_SECRET_KEY=[redacted]

# -------------------------
# Paperless Basics
# -------------------------
PAPERLESS_TIME_ZONE=Europe/Berlin
PAPERLESS_OCR_LANGUAGE=deu+eng

# -------------------------
# Redis
# -------------------------
PAPERLESS_REDIS=redis://broker:6379

# -------------------------
# PostgreSQL 18
# -------------------------
POSTGRES_DB=paperless
POSTGRES_USER=paperless
POSTGRES_PASSWORD=[redacted]
# PG18+: Datenverzeichnis im Volume
PGDATA=/var/lib/postgresql/18/docker

# Paperless DB-Verbindung
PAPERLESS_DBHOST=db
PAPERLESS_DBNAME=paperless
PAPERLESS_DBUSER=paperless
PAPERLESS_DBPASS=[redacted]

# -------------------------
# Paperless Optional Settings
# -------------------------
PAPERLESS_FILENAME_FORMAT={{ created_year }}/{{ created }}-{{ correspondent }}-{{ document_type }}-{{ title }}
PAPERLESS_FILENAME_FORMAT_REMOVE_NONE=true
PAPERLESS_FILENAME_DATE_ORDER=YMD
PAPERLESS_CONSUMER_DELETE_DUPLICATES=true
PAPERLESS_CONSUMER_RECURSIVE=true
PAPERLESS_IGNORE_DATES=1970-01-01
PAPERLESS_OCR_USER_ARGS={"invalidate_digital_signatures":true}
```

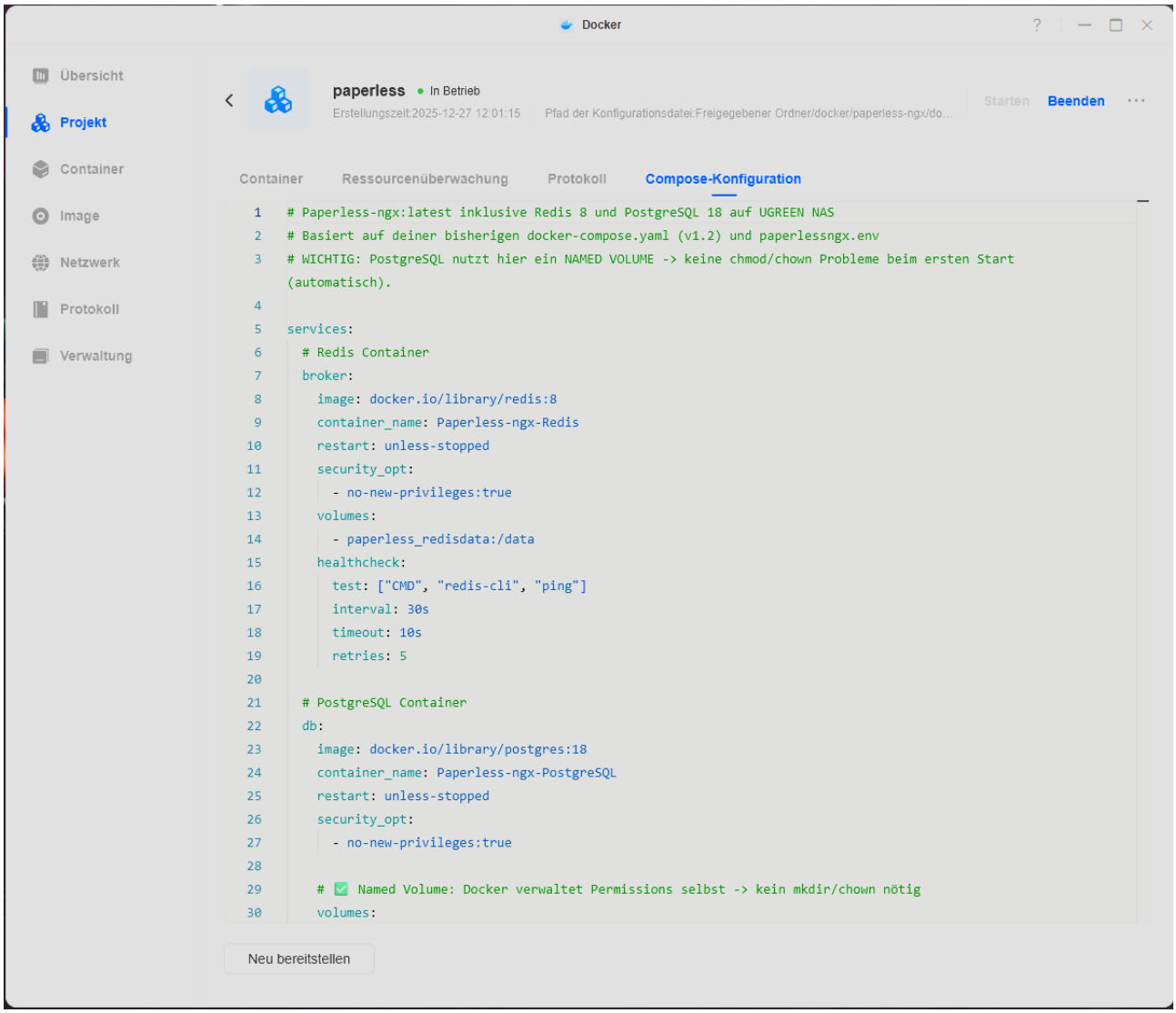
Normal text file Länge: 3.319 Zeilen: 102 Zei: 1 Spa: 1 Pos: 1 Unix (LF) UTF-8 INS

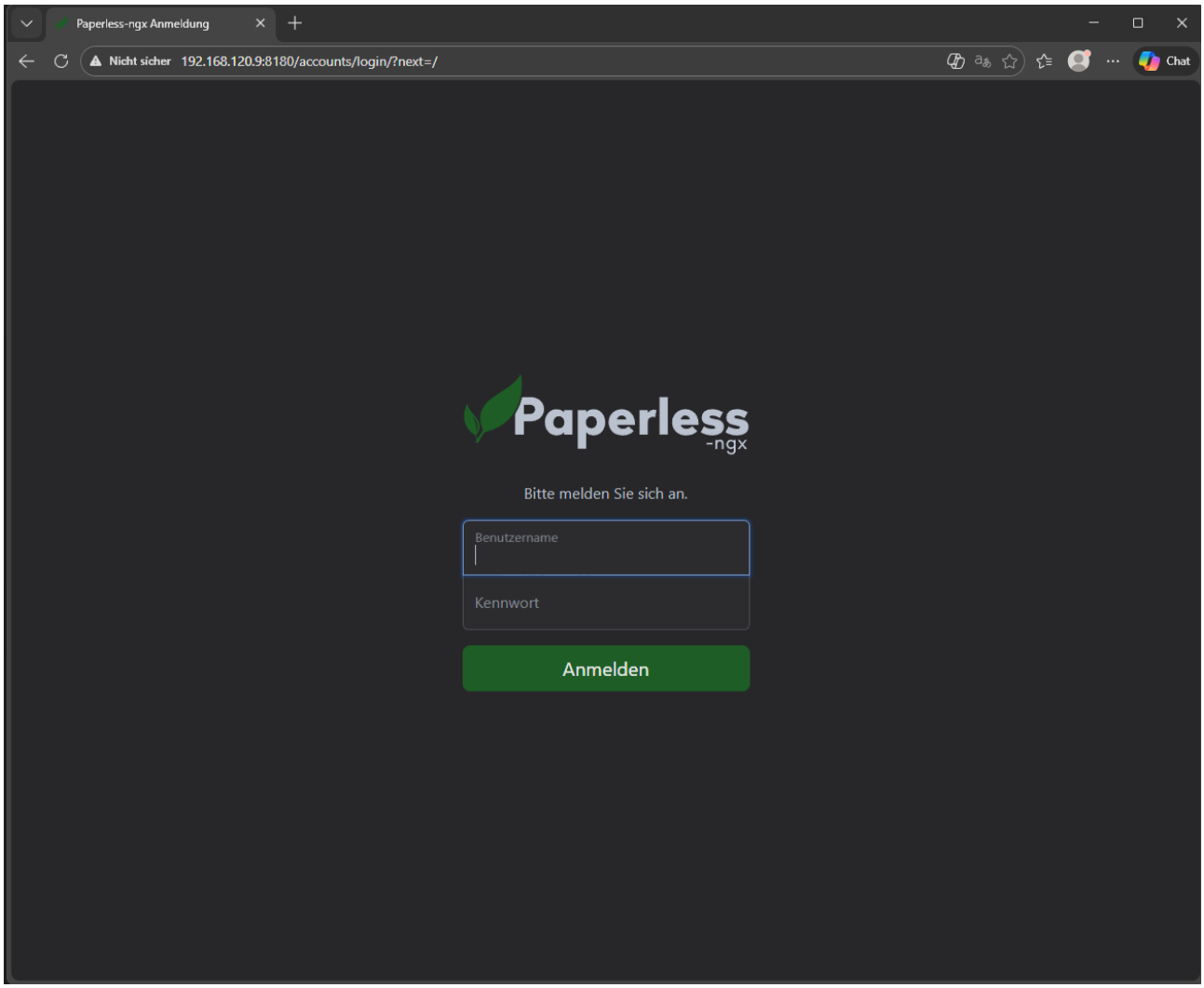
5.3 Import and start the Docker project

In UGOS: Docker -> Projects -> Import -> select docker-compose.yaml. Then start the project. The web interface will be available on port 8180:

```
http://<NAS-IP>:8180
```

Note: If your Docker Compose environment does not support “depends_on: condition: service_healthy”, you can change the depends_on block in docker-compose.yaml to the simple variant (see comment in the file).





6. Configuration (paperlessngx.env)

6.1 Database password (important)

POSTGRES_PASSWORD and PAPERLESS_DBPASS must match exactly. If you change the password after the first start, it will not be applied to the existing database automatically.

6.2 SMTP / email for the export script

The following variables control email sending:

```
MAIL_ENABLED=1
MAIL_ON_SUCCESS=0
MAIL_ON_FAILURE=1
MAIL_ASCII_ICONS=0
```

Optional: Send email as multipart (text + HTML):

```
MAIL_SEND_HTML=1
```

Optional: Backup list in the email (latest ZIP backups in the export folder):

```
MAIL_FILELIST_ENABLED=1
MAIL_FILELIST_LIMIT=100
```

Optional: Exported files (ZIP contents) in the email:

```
MAIL_EXPORTED_FILES_ENABLED=1
MAIL_EXPORTED_FILES_LIMIT=50
MAIL_EXPORTED_FILES_MODE=primary
MAIL_EXPORTED_FILES_PRIMARY_EXTS=pdf
```

Tip: For very large exports, disable the exported file list to keep the email smaller.

SMTP settings:

```

SMTP_HOST=smtp.example.com
TLS_ENABLED=1
SMTP_PORT=
SMTP_USER=user@example.com
SMTP_PASS="CHANGE_ME_SMTP_PASSWORD"

MAIL_FROM=user@example.com
MAIL_FROM_NAME="UGREEN Paperless Export"
MAIL_TO=target@example.com
```

Note: Multiple recipients are supported (comma-separated), e.g. MAIL_TO=a@b.com,b@c.com

6.3 Export target and retention

The export script writes ZIP files to the host target folder:

```
PAPERLESS_HOST_BACKUP_DIR=/volume2/docker/paperless-ngx/export
PAPERLESS_KEEP_FILES=10
PAPERLESS_CONTAINER_TMP_DIR=/usr/src/paperless/export/_tmp
```

Optional: Logfile (host path). If empty, the script uses ./scripts/exporter.log:

```
PAPERLESS_EXPORT_LOGFILE=/volume2/docker/paperless-ngx/scripts/exporter.log
```

Note: If you really want to use /tmp, set PAPERLESS_CONTAINER_TMP_DIR=/tmp/paperless-export. The script will try to chmod 777 inside the container (docker exec -u 0).

7. Test the export script

7.1 Make the script executable

SSH note: Run this as root. If you logged in via SSH, use `sudo -i` first.

```
chmod +x /volume2/docker/paperless-ngx/scripts/paperless-export_with_mail.sh
```

7.2 Start the export manually

```
/volume2/docker/paperless-ngx/scripts/paperless-export_with_mail.sh
```

The script creates a file like paperless_export_YYYYmmdd_HHMMSS.zip in the export folder and writes a logfile (default: scripts/exporter.log, configurable via PAPERLESS_EXPORT_LOGFILE).

Tip: To test email, temporarily set MAIL_ON_SUCCESS=1 and run the script manually.

Placeholder: Screenshot 6 – exporter.log or the export ZIP in the export/ folder.

8. Automation with cron (optional)

Cron jobs should be set up as root (SSH). SSH note: run `sudo -i` first. Open crontab:

```
crontab -e
```

Example: daily export at 02:10:

```
10 2 * * * /volume2/docker/paperless-ngx/scripts/paperless-export_with_mail.sh
>/dev/null 2>&1
```

9. Updates / maintenance

- Update Docker images (UGOS project: Stop -> Pull/Update -> Start).
- Before major updates, create an export backup (run the script manually).
- Keep paperlessngx.env safe (contains passwords).

10. Troubleshooting

10.1 Database login failed (password authentication failed)

If the webserver cannot connect to the database, check whether POSTGRES_PASSWORD and PAPERLESS_DBPASS are identical and whether the database volume already contains a database. If you changed passwords after the first start, you usually need to remove the database volume (this deletes the database) or adjust credentials accordingly.

```
# Stop the project, then delete the volume (example)
docker volume ls | grep -i paperless_pgdata
docker volume rm <VOLUME_NAME>
```

10.2 Check container logs

```
docker logs --tail 200 Paperless-ngx-PostgreSQL
docker logs --tail 200 Paperless-ngx
docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}'
```

11. Notes & liability

This is a community solution and not an official UGREEN product. Use at your own risk. I assume no liability for data loss or damage. Always test backups and keep additional backups of important data.